OSI Model -- Guided Notes ANSWER KEY

Unit 1.1.1 | CompTIA Network+ N10-009 Obj. 1.1 | N10-008 1.1

For instructor use / distribute after students complete the notes.

Question 1

The OSI model divides network communication into _______ layers, from the _______ layer at the bottom to the _______ layer at the top.

Answer: Seven (7) layers -- Physical at the bottom, Application at the top.

Question 2

Layer 1 devices -- such as hubs, repeaters, and _______ -- deal only with raw _______ and have no understanding of addresses or protocols.

Answer: Network interface cards (NICs); raw bits (electrical signals, light pulses, or radio waves).

Question 3

A switch operates at Layer _______ and uses _______ addresses to forward frames to the correct port, rather than flooding traffic to every device like a hub.

Answer: Layer 2 (Data Link); MAC addresses.

Question 4

Explain the difference between a MAC address and an IP address. Which OSI layer is each one associated with?

Answer: A MAC address is a hardware address used at Layer 2 (Data Link) to deliver frames on a local network. An IP address is a logical address used at Layer 3 (Network) to route packets across different networks. MAC = local delivery; IP = internet routing.

Question 5

At Layer 4, the two main transport protocols are _______ and _______. The first provides reliable, ordered delivery; the second prioritizes _______ over reliability.

Answer: TCP (Transmission Control Protocol) and UDP (User Datagram Protocol); speed.

Question 6

Port numbers exist at Layer _______. They allow a single device with one IP address to run multiple _______ simultaneously (for example, a web server and an SSH daemon on the same machine).

Answer: Layer 4 (Transport); services (or applications).

Question 7

When a browser negotiates a TLS connection to load an HTTPS website, which OSI layer handles that encryption setup? _______

Answer: Layer 6 (Presentation).

Question 8 -- Real World Application

REAL WORLD: A user reports that they can ping 8.8.8.8 successfully but cannot load any websites in their browser. Based on the OSI model, which layer(s) would you investigate first, and what specific service is most likely the cause? Explain your reasoning in two to three sentences.

Answer: Layers 1-3 are working (ping to external IP succeeds). The problem is at Layer 7 (Application) -- specifically DNS. The computer can reach the internet by IP but cannot resolve domain names. Check DNS configuration with nslookup; correct or replace the DNS server address.